HRP · FOUR-ASSET WALKTHROUGH

Hierarchical Risk Parity

A step-by-step four-asset example

Following the MathWorks HRP example from correlations to final weights

Tree clustering → quasi-diagonalization → recursive bisection, computed by hand, then rebuilt with Claude Code.

HRP splits risk along a tree, without inversion

Hierarchical Risk Parity was proposed by Marcos López de Prado (2016).

1

Why: mean-variance optimization is fragile

It inverts the covariance matrix, so small estimation errors push weight into a few assets.

2

How: order first, then split the budget

Tree clustering and quasi-diagonalization set the order; recursive bisection sets the weights.

3

Result: every asset gets a sensible share

Our example ends at A1 0.258, A2 0.237, A3 0.257, A4 0.248.

Every number here is computed by hand from the MathWorks example

Six parts take us from basics to a hands-on build

Parts 2 to 4 are the three stages of HRP; the same four-asset example runs throughout.

Part	Topic	What you will see
01	Basics	Why MVO concentrates, inverse-variance weights, clustering
02	Tree clustering	Correlations → distances → dendrogram
03	Quasi-diagonalization	Reorder the assets by the leaf order
04	Recursive bisection	Split in half, again and again, to final weights
05	Results	A what-if, the 14-asset result, strengths and limits
06	Build	Rebuild everything with Claude Code prompts

01

PART 1 · BASICS

Background for Reading HRP

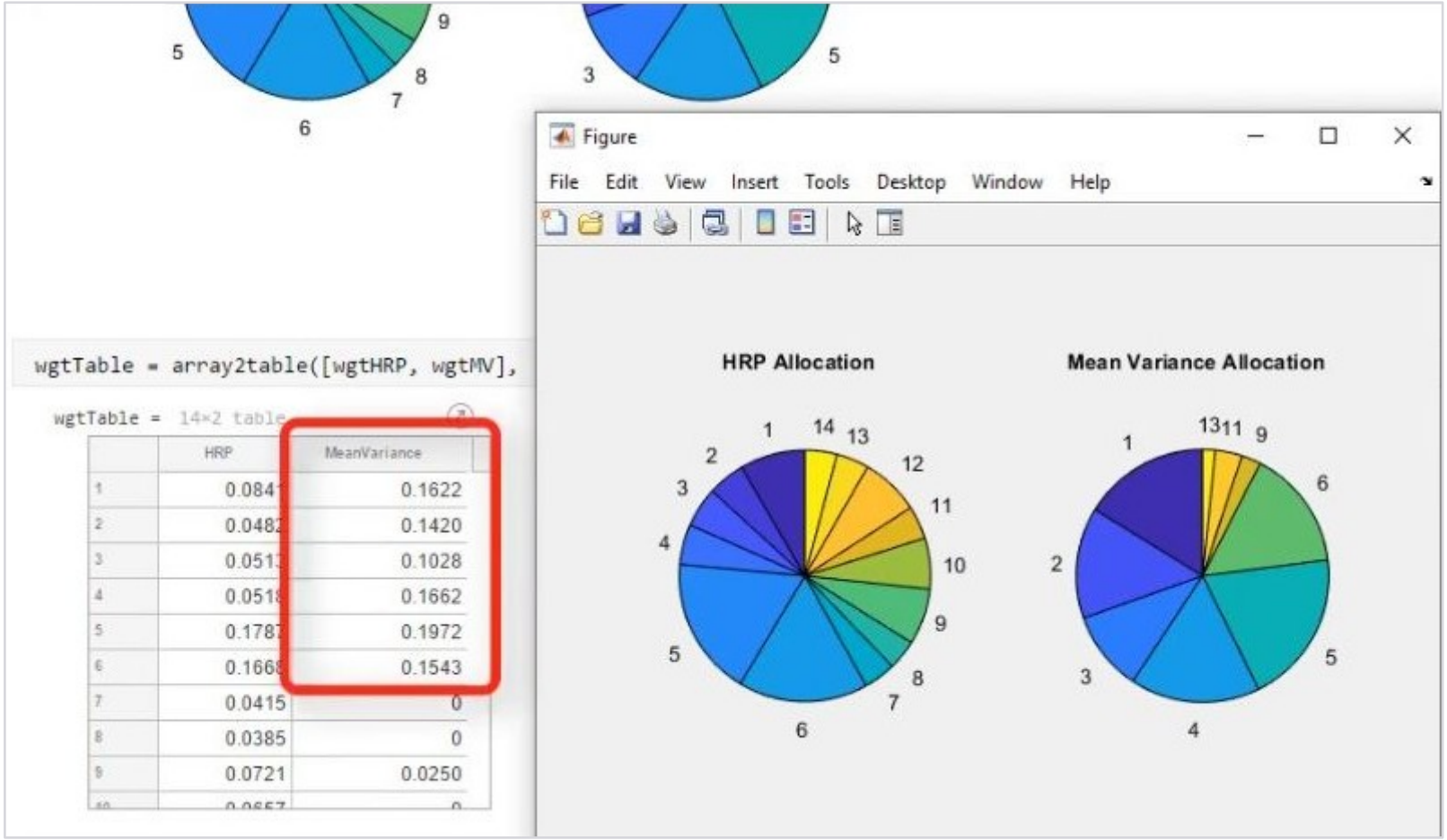
The weakness of mean-variance optimization, inverse-variance weights and hierarchical clustering.

Mean-variance piles weight into a few assets

Original MathWorks screen: the same 14 assets allocated by HRP and by mean-variance (MVO).

Why it happens

- MVO inverts the covariance matrix
- Correlated assets make it unstable
- Small errors → big weight swings
- Assets 7 and 8 get zero



Inverse variance gives the calmer asset more

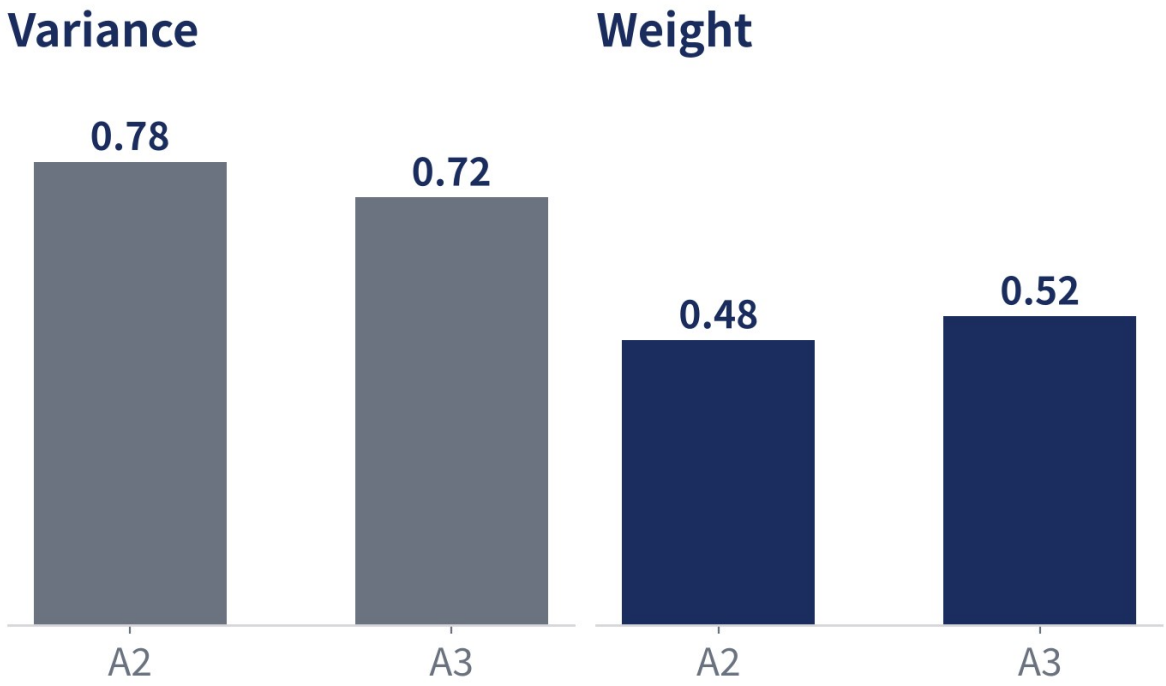
HRP uses this rule inside each group. Variance is the diagonal of the covariance matrix.

Weight rule

$$w_i = \frac{1/\mathrm{Var}_i}{\sum_j 1/\mathrm{Var}_j}$$

Why this rule

- Lowest variance if assets were uncorrelated
- HRP handles correlation with the tree



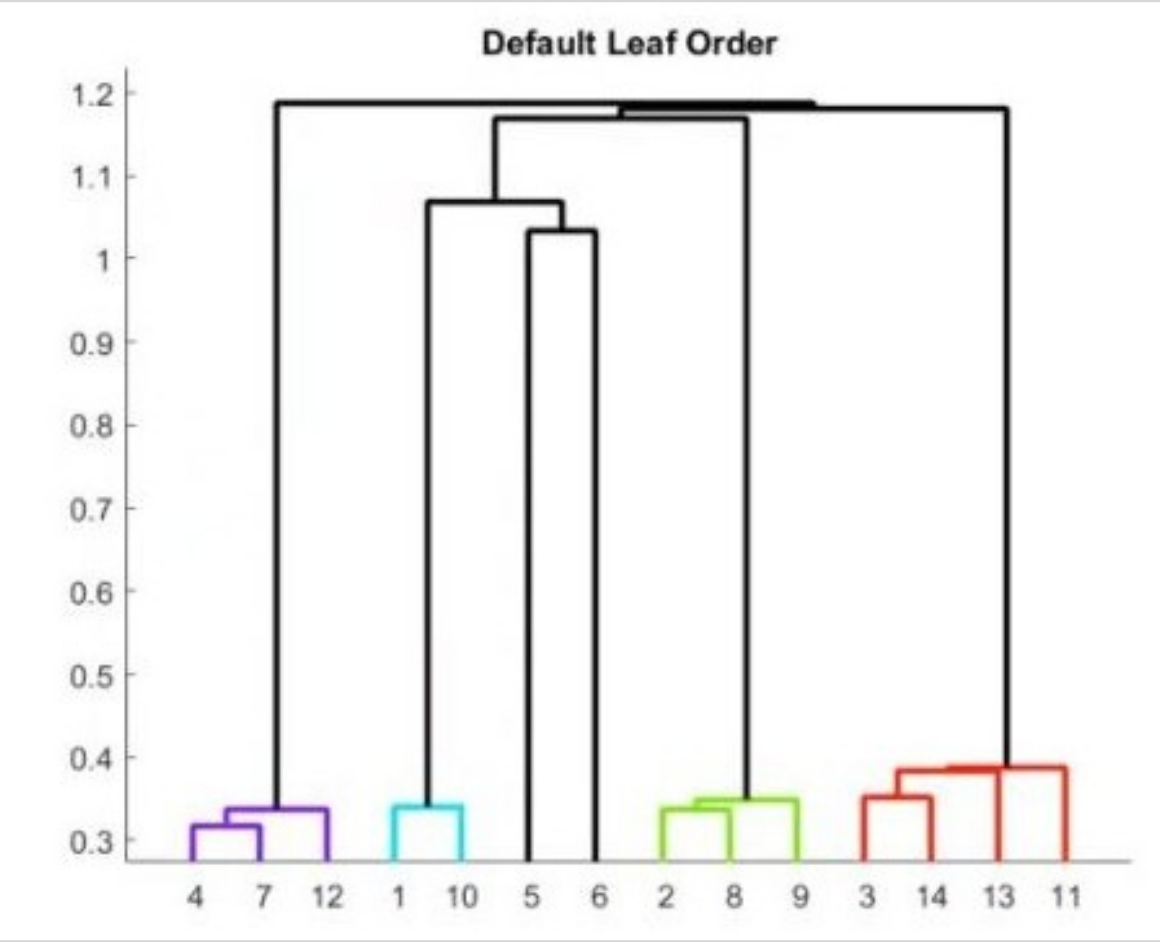
A3 has the smaller variance (0.72), so it gets the larger weight (0.52)

Hierarchical clustering joins the closest assets first

Dendrogram of the original 14-asset MathWorks example.

Four words

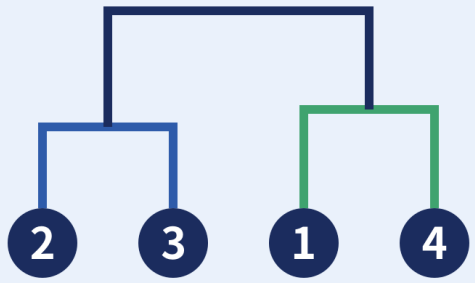
- Distance: how different
- Linkage: group distance rule
- Dendrogram: the merge tree
- Leaf: one asset at the bottom



HRP works in three stages, from ordering to weighting

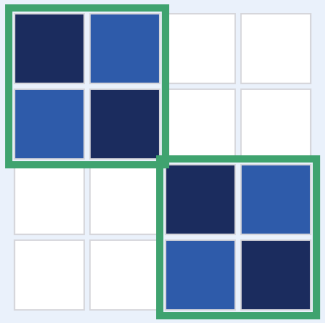
Stages 1 and 2 only decide the order of the assets; stage 3 turns that order into weights.

1 Tree clustering



- correlation → distance
- merge closest pairs
- draw a dendrogram

2 Quasi-diagonalization



- read the leaf order
- reorder rows, columns
- similar assets together

3 Recursive bisection



- cut the list in half
- safer half gets more
- repeat to each asset

The example has four assets that form two pairs

Made-up numbers chosen so that the MathWorks animation values come out exactly.

Correlation — stages 1 and 2

	A1	A2	A3	A4
A1	1.00	0.07	0.07	0.26
A2	0.07	1.00	0.31	0.06
A3	0.07	0.31	1.00	0.07
A4	0.26	0.06	0.07	1.00

Covariance — stage 3

	A1	A2	A3	A4
A1	0.74	0.05	0.05	0.20
A2	0.05	0.78	0.23	0.05
A3	0.05	0.23	0.72	0.05
A4	0.20	0.05	0.05	0.77

Close pairs: A2–A3 (corr 0.31) and A1–A4 (0.26). The covariance diagonal holds each asset's variance.

Correlation across the two pairs is only about 0.07

02

PART 2 · CLUSTERING

From Correlations to a Dendrogram

What three lines of MathWorks code do, followed with four-asset numbers.

A higher correlation means a shorter distance

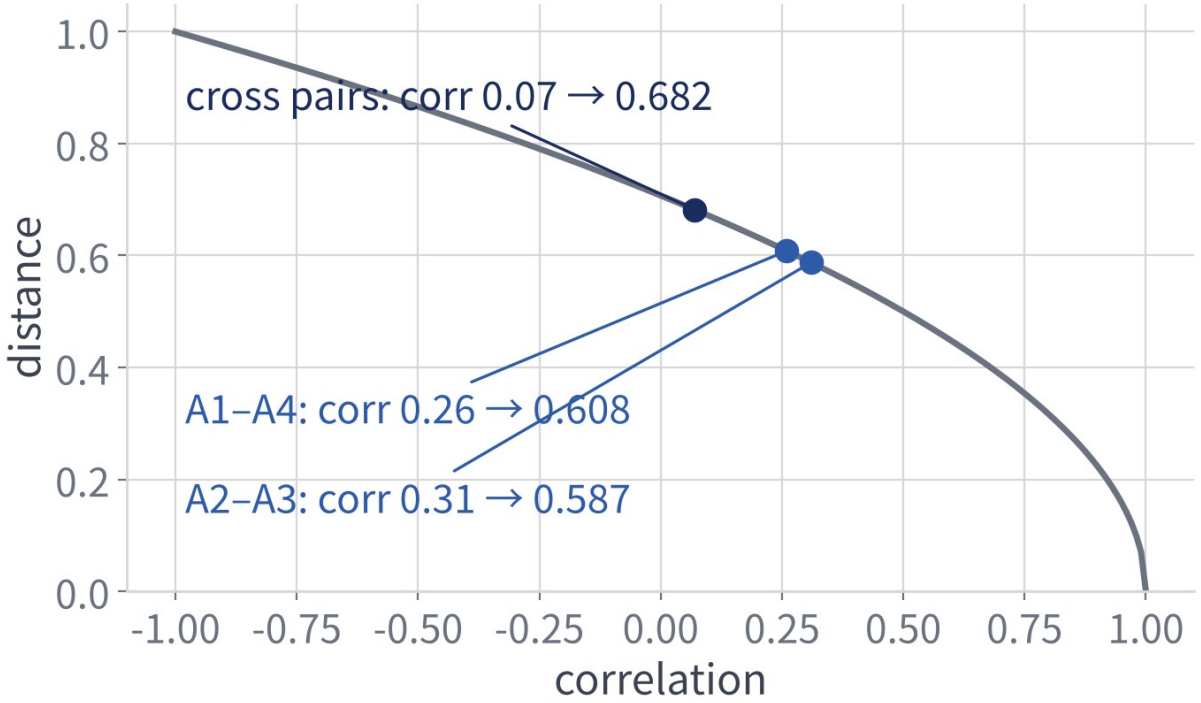
First line of the original code. Correlation 1 gives distance 0; correlation −1 gives distance 1.

Correlation distance

$$\text{distance} = \sqrt{\frac{1-\text{corr}}{2}}$$

MATLAB · original code, line 10

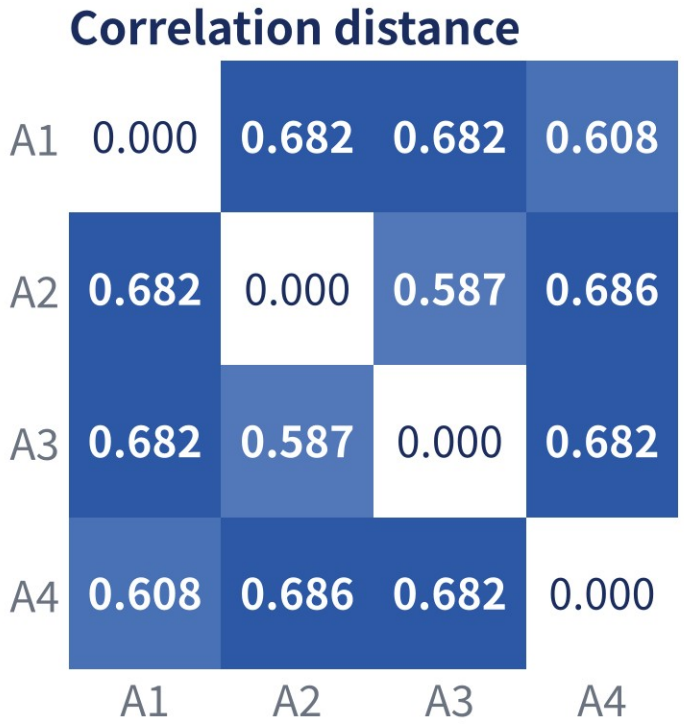
```
distCorr = ((1-cr)/2).^0.5;
```



The two close pairs land at 0.587 and 0.608; every cross pair sits near 0.68

The two pairs have the smallest distances

Light cells are close pairs. The diagonal is each asset's distance to itself.



How to read it

- A2–A3 is the smallest: 0.587
- A1–A4 comes next: 0.608
- All other pairs: about 0.68
- Next, each row becomes the asset's coordinates

This table already suggests that A2–A3 and A1–A4 will merge first

MATLAB then measures the distance between rows

The 'euclidean' option treats each asset's row as a point with four coordinates.

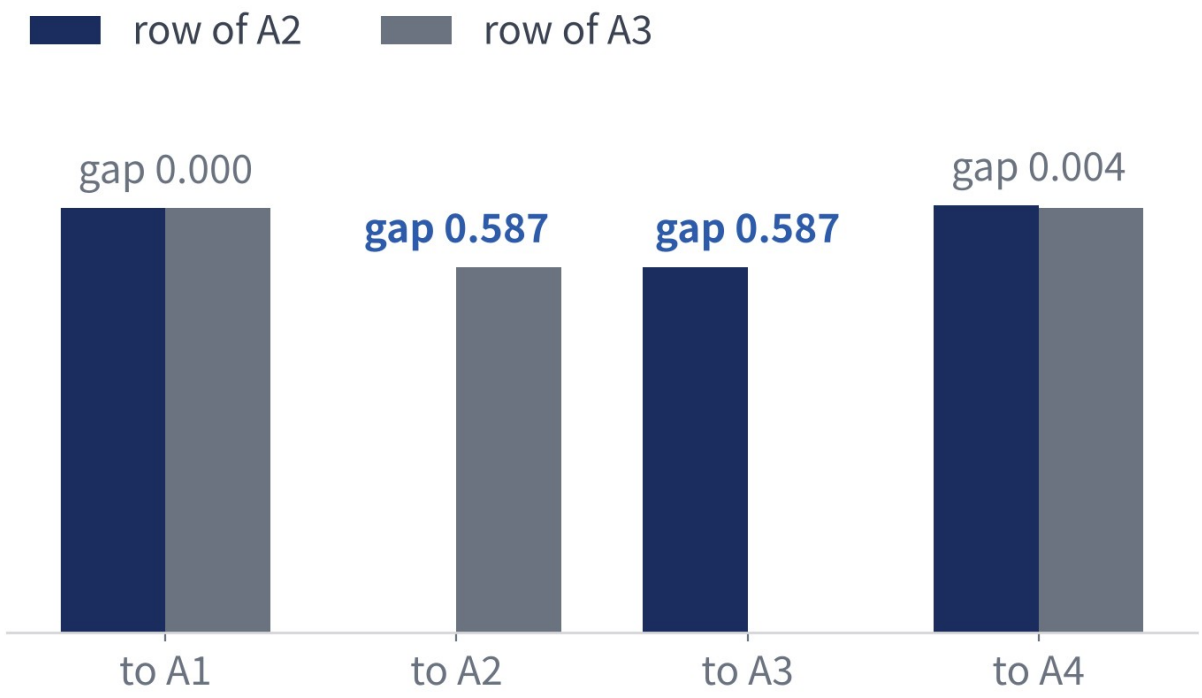
MATLAB · original code, line 13

```
link = linkage(distCorr,
       'single', 'euclidean');
```

A2 vs A3: two gaps are not zero

$$\sqrt{0.587^2 + 0.587^2} \approx 0.830$$

Rows that look alike are close.



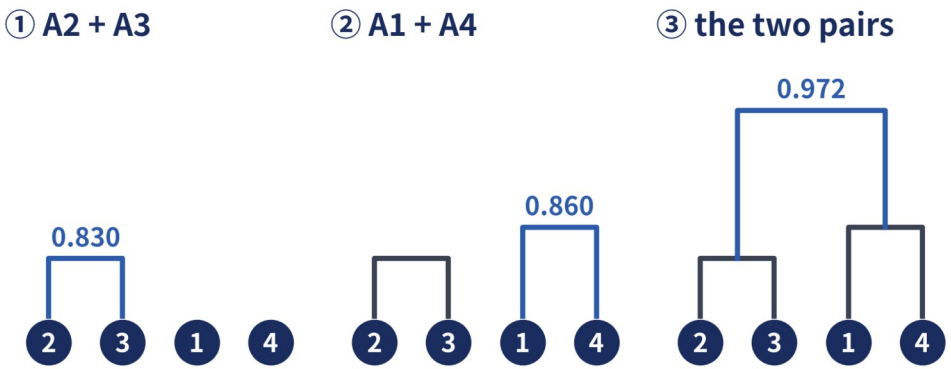
A2 and A3 differ only in their own two columns, so their rows end up 0.830 apart

Single linkage merges the closest pair, then the next

Between two groups, single linkage uses the distance of their two closest members.

Distance between rows

	A1	A2	A3	A4
A1	0.000	0.972	0.972	0.860
A2	0.972	0.000	0.830	0.978
A3	0.972	0.830	0.000	0.972
A4	0.860	0.978	0.972	0.000

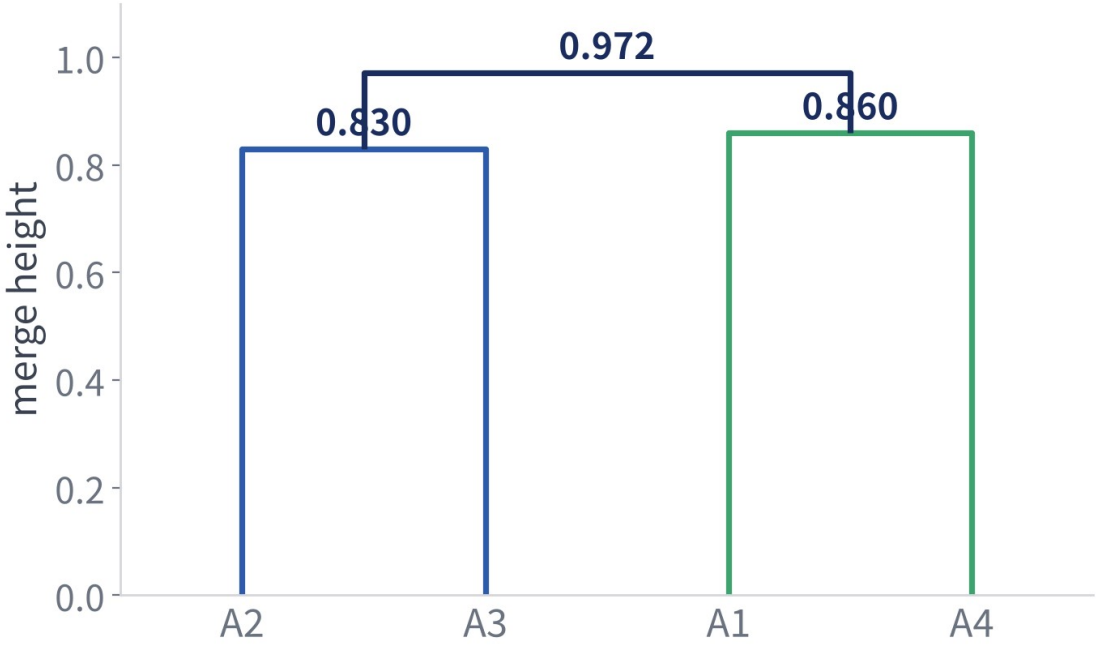


merge	with	height	new name
A2	A3	0.830	group 5
A1	A4	0.860	group 6
group 5	group 6	0.972	group 7 (all)

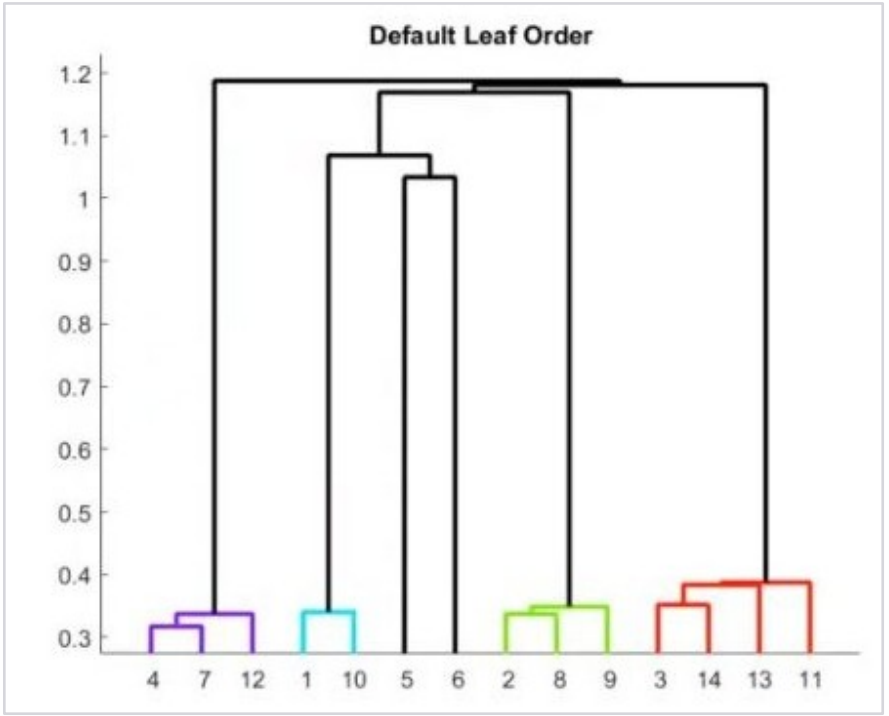
Step ③ uses 0.972, the smallest of the four cross-pair distances

The dendrogram shows two pairs that meet at the top

Left: our four-asset example. Right: the original 14-asset example.



Four assets (numbers are merge heights)



Original 14 assets

Four assets meet at 0.972; 14 assets form four groups plus two loners

03

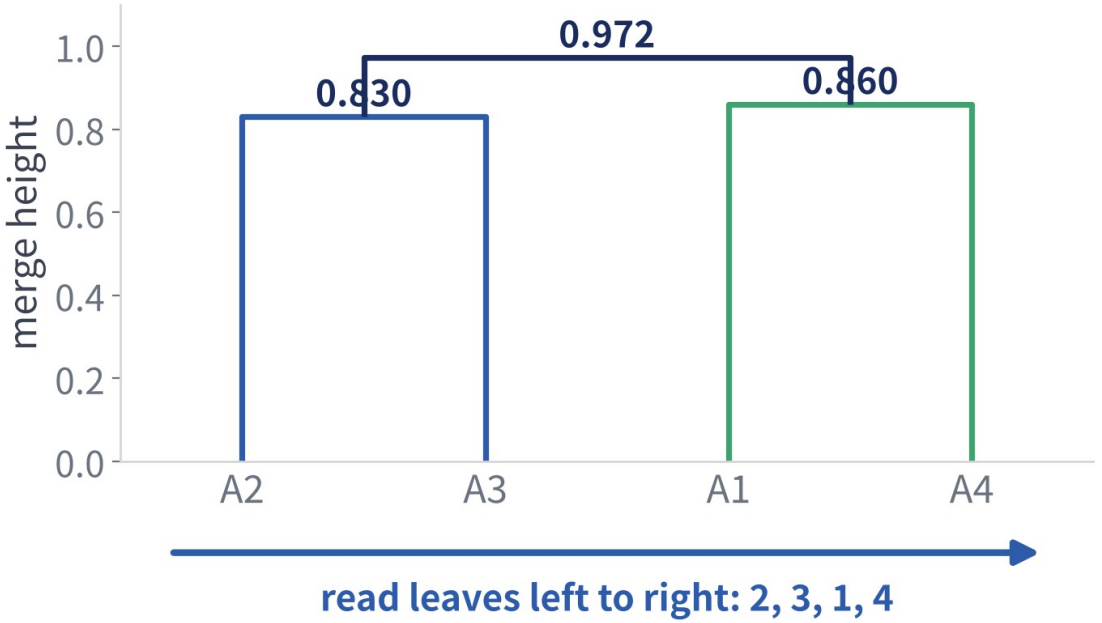
PART 3 · QUASI-DIAG

Reordering by the Dendrogram

Read the leaves from left to right and line the assets up in that order.

Reading the leaves gives the order 2, 3, 1, 4

The original code collects the leaves under the root group, from left to right.



MATLAB · original code, lines 17–19

```
numLeafNodes = size(link,1)+1;
rootGroupNodeId = 2*numLeafNodes-1;
sortedIdx = getLeafNodesInGroup(...);
```

Inside the root (group 7)

- Left branch: group 5 = A2, A3
- Right branch: group 6 = A1, A4

sortedIdx is 2, 3, 1, 4, and stage 3 uses nothing but this order

Reordering gathers similar assets on the diagonal

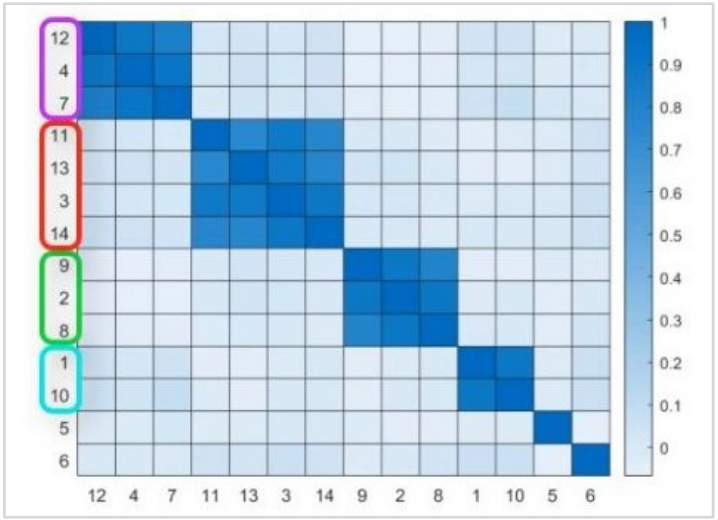
Left: the four-asset example before and after. Right: the original 14-asset result.

Original order 1, 2, 3, 4

	A1	A2	A3	A4
A1	1.00	0.07	0.07	0.26
A2	0.07	1.00	0.31	0.06
A3	0.07	0.31	1.00	0.07
A4	0.26	0.06	0.07	1.00

Leaf order 2, 3, 1, 4

	A2	A3	A1	A4
A2	1.00	0.31	0.07	0.06
A3	0.31	1.00	0.07	0.07
A1	0.07	0.07	1.00	0.26
A4	0.06	0.07	0.26	1.00



Original 14 assets

No number changes, only the order — and the dark cells gather into diagonal blocks

04

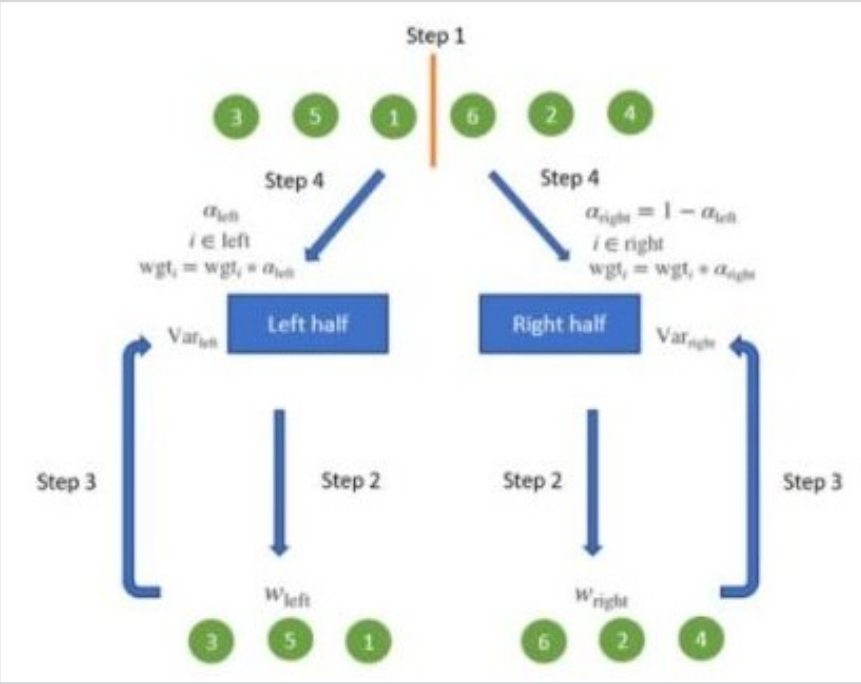
PART 4 · BISECTION

Splitting the Budget in Halves

The four moves of the MathWorks animation, computed to the end with our numbers.

Bisection repeats four moves down to each asset

Steps 1 to 4 in the original flow chart; they all happen inside HRP stage 3.

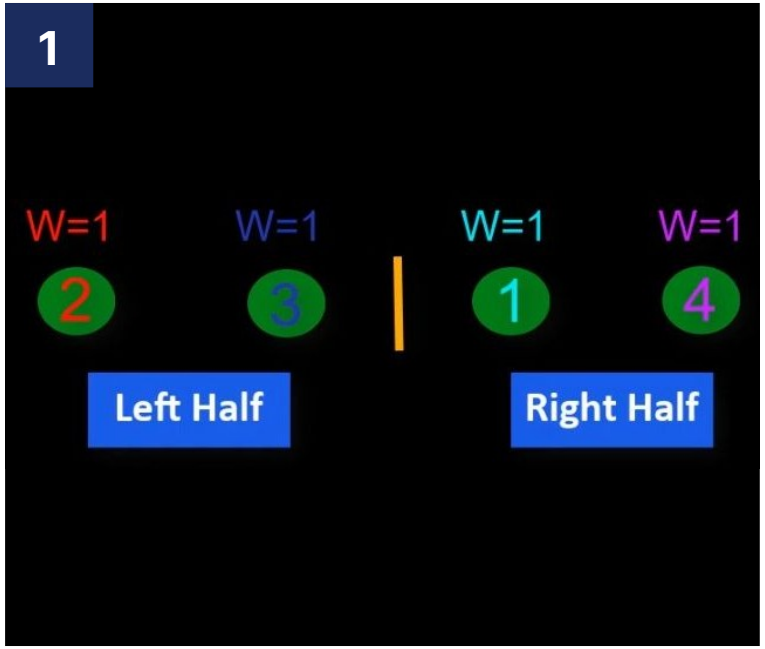


Step	What happens
1 · Split	Cut the list in two
2 · Temp weights	Inverse-variance weights
3 · Half variance	Variance of each half
4 · Allocate	Safer half gets more

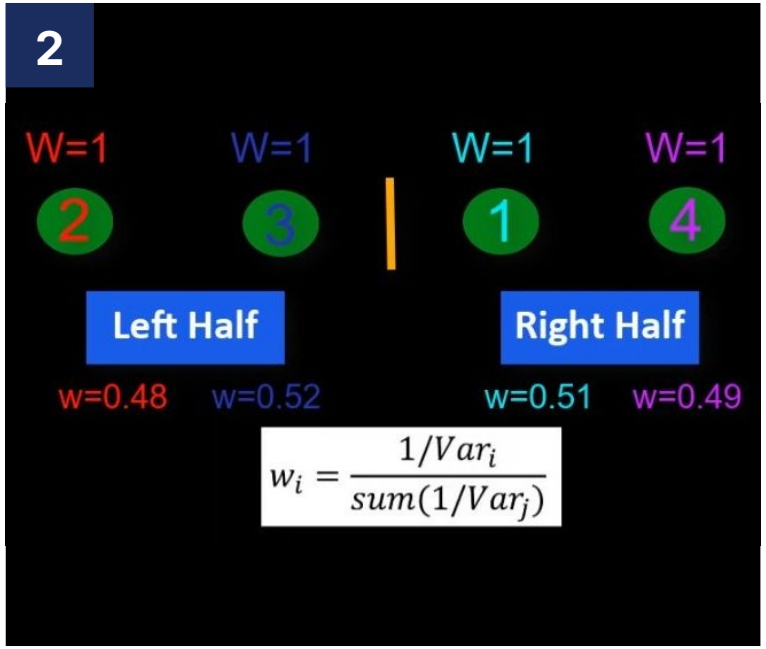
Repeat Steps 1 to 4 until every half holds a single asset

The animation first splits and weighs each half

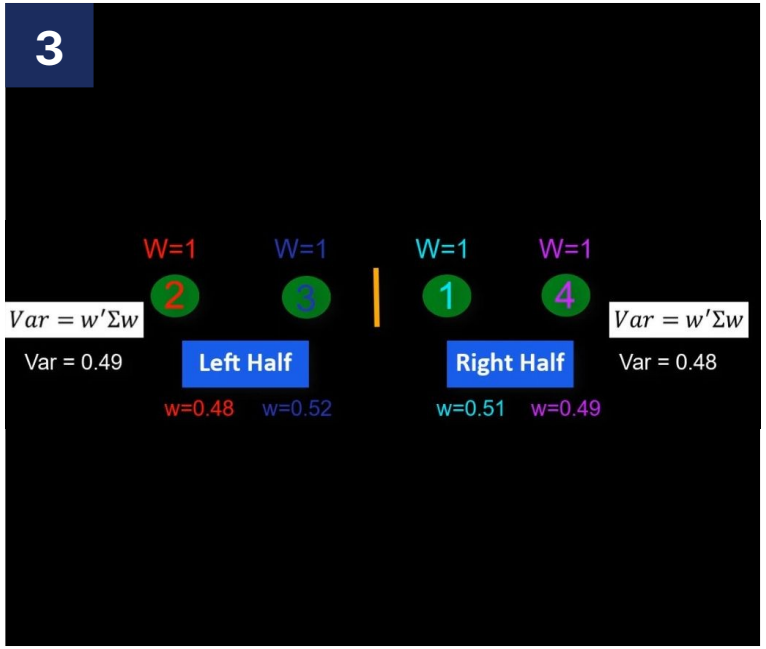
Frames 1–3 of the MathWorks video: split, temporary weights, half variances.



Split: every budget W starts at 1



Temporary weights w per half

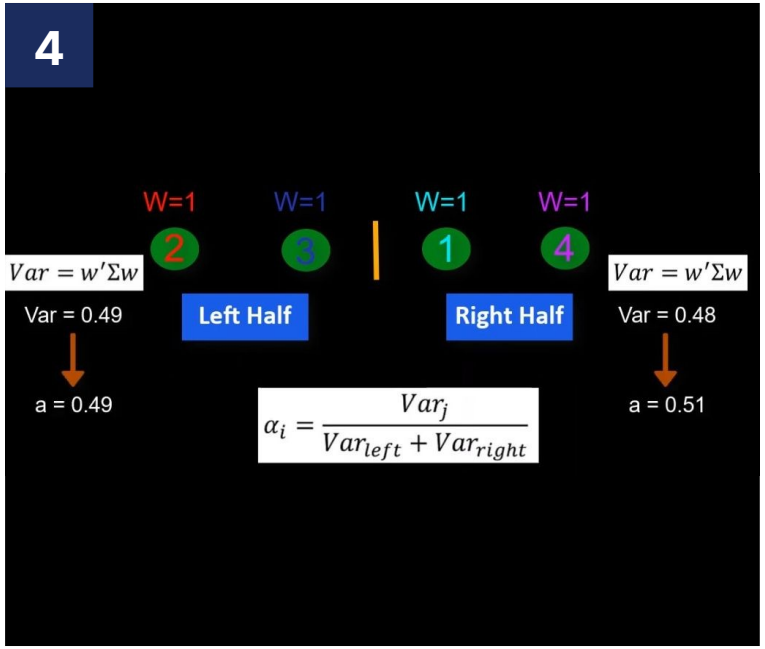


Half variances 0.49 and 0.48

Frames 1–3 match Steps 1–3, redrawn with exact numbers next

It then turns variances into budgets

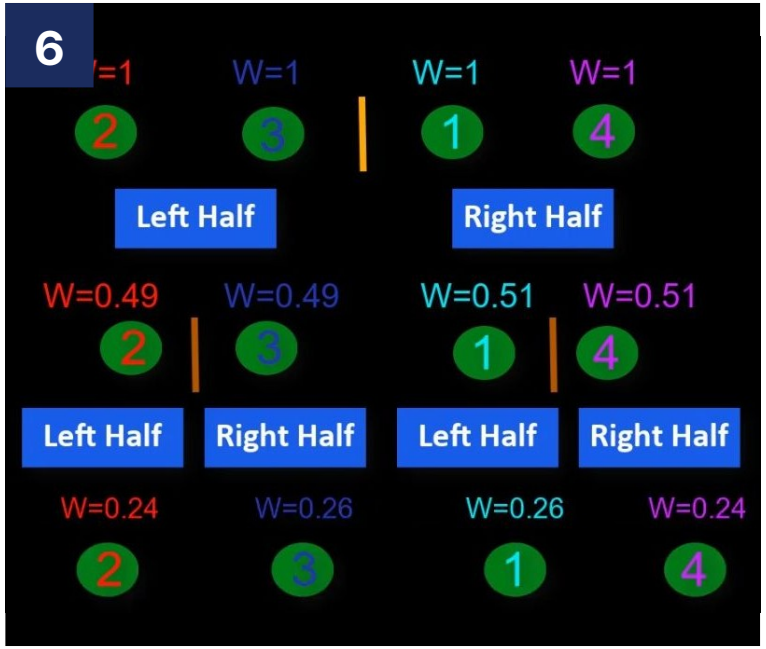
Frames 4–6: split ratios, new budgets, and the second split to final weights.



Split ratios a = 0.49 and 0.51



New budgets W after level one

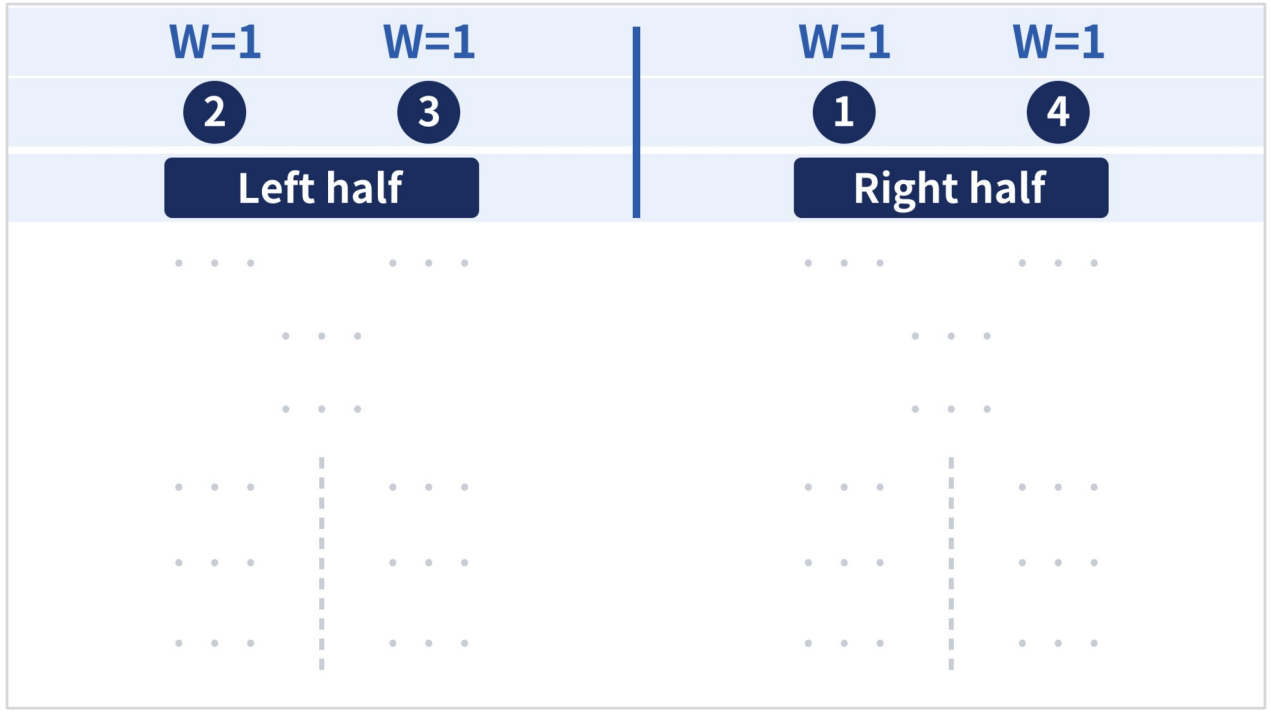


Second split gives final weights

Frames 4–6 match Step 4, the new budgets and the final weights

Every asset starts at 1, then the list is halved

W is the budget each asset holds. Grey dots are values still to be computed.



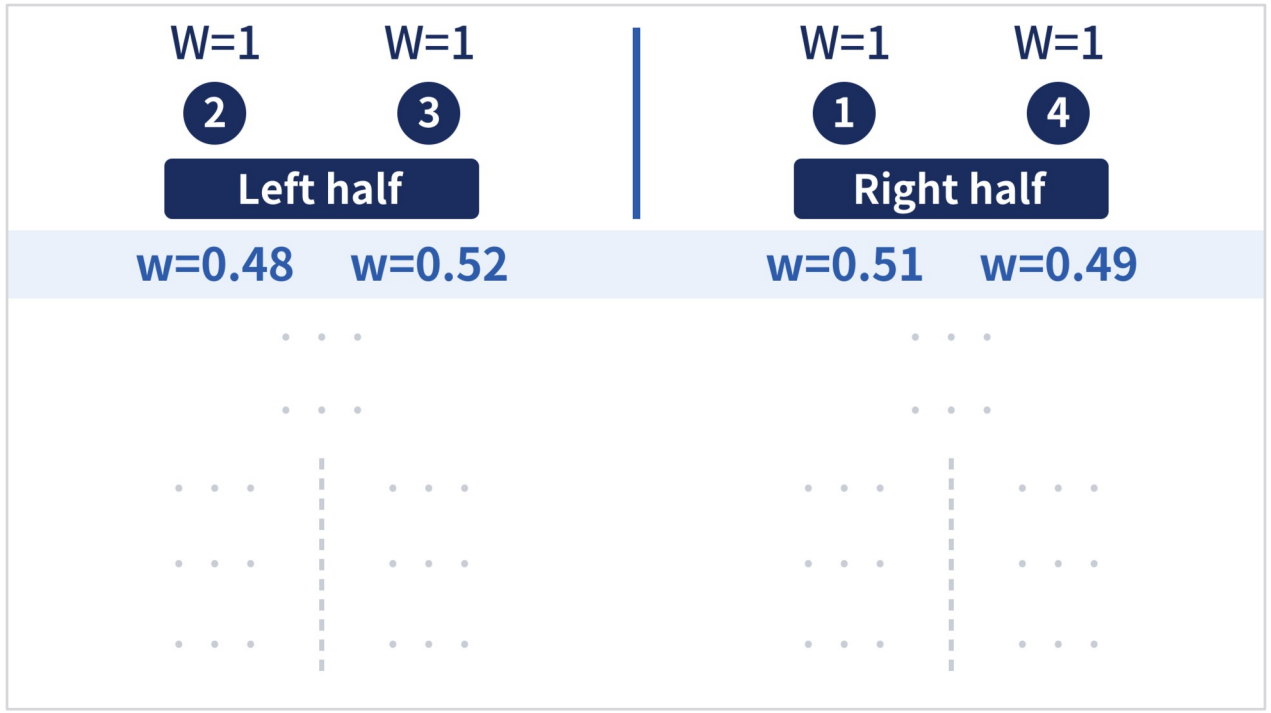
What happens here

- Cut the order 2, 3, 1, 4 in the middle
- Left half: A2, A3
- Right half: A1, A4
- The cut is at half the count

Here the halfway cut happens to fall exactly between the two pairs

Inverse variance sets temporary weights per half

The same rule as in Basics, applied to each half separately.



Rule

$$w_i = \frac{1/\mathrm{Var}_i}{\sum_j 1/\mathrm{Var}_j}$$

Our numbers (A3 = 0.52, A4 = 0.49)

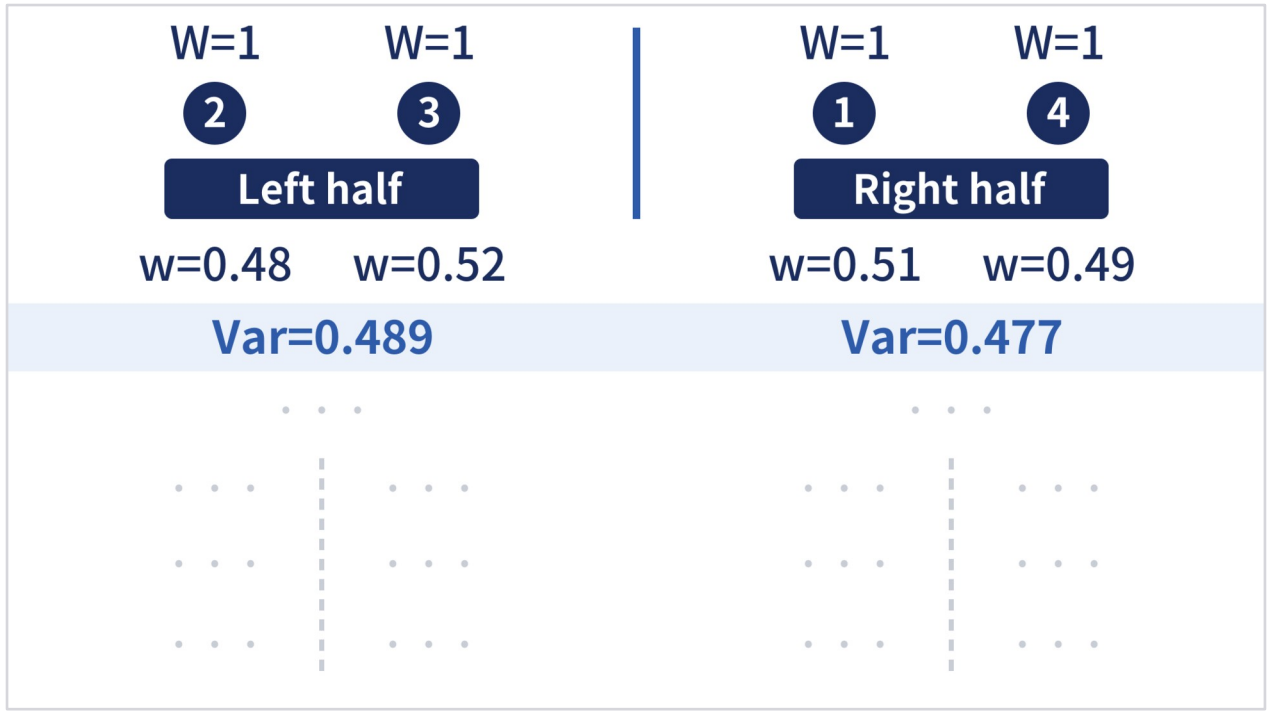
$$w_2 = \frac{1/0.78}{1/0.78 + 1/0.72} = 0.48$$

$$w_1 = \frac{1/0.74}{1/0.74 + 1/0.77} = 0.51$$

These match the animation's w = 0.48, 0.52, 0.51, 0.49

The temporary weights give each half its variance

Treat each half as a small portfolio. Only covariances inside the half are used.



Rule

$$\mathrm{Var} = w' \Sigma w$$

Our numbers

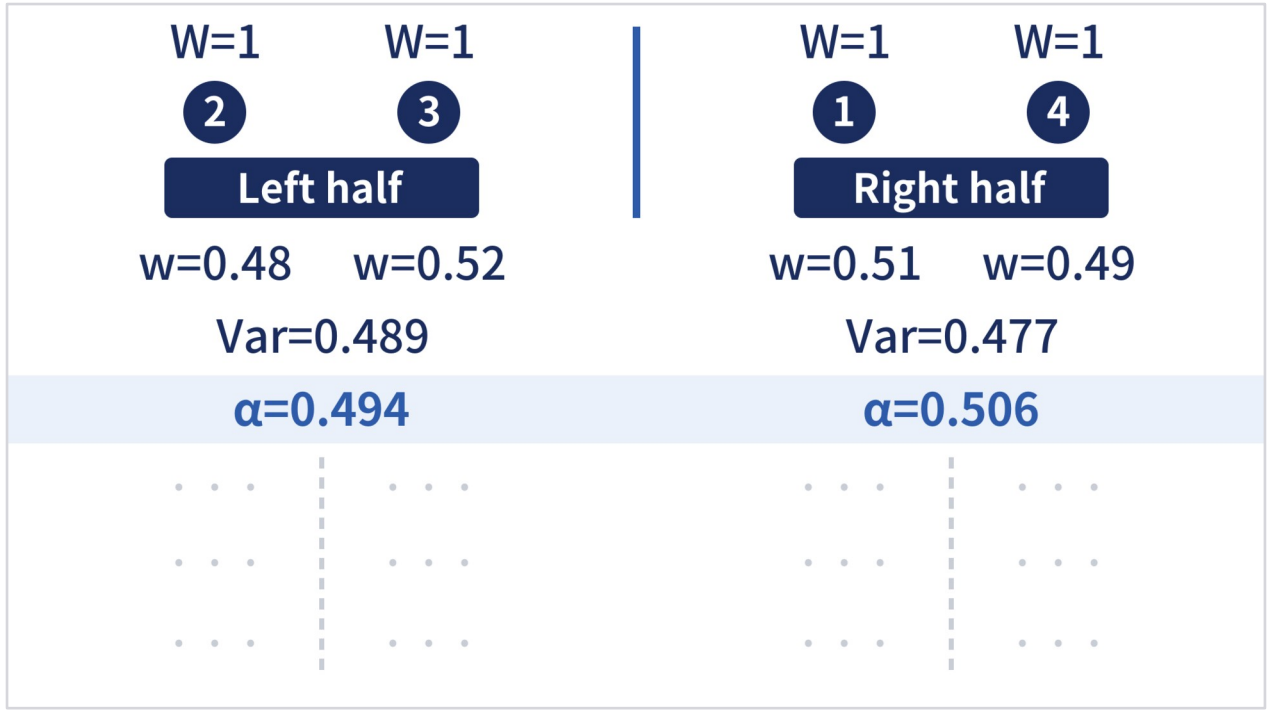
$$\mathrm{Var}_{\mathrm{left}} = 0.48^2(0.78) + 0.52^2(0.72) + 2(0.48)(0.52)(0.23) = 0.489$$

$$\mathrm{Var}_{\mathrm{right}} = 0.51^2(0.74) + 0.49^2(0.77) + 2(0.51)(0.49)(0.20) = 0.477$$

The left half (0.489) is slightly riskier than the right half (0.477)

The safer half receives the larger share

The other half's variance goes on top of the fraction. The animation calls it a.



Rule

$$\alpha_{\text{left}} = \frac{\text{Var}_{\text{right}}}{\text{Var}_{\text{left}} + \text{Var}_{\text{right}}}$$

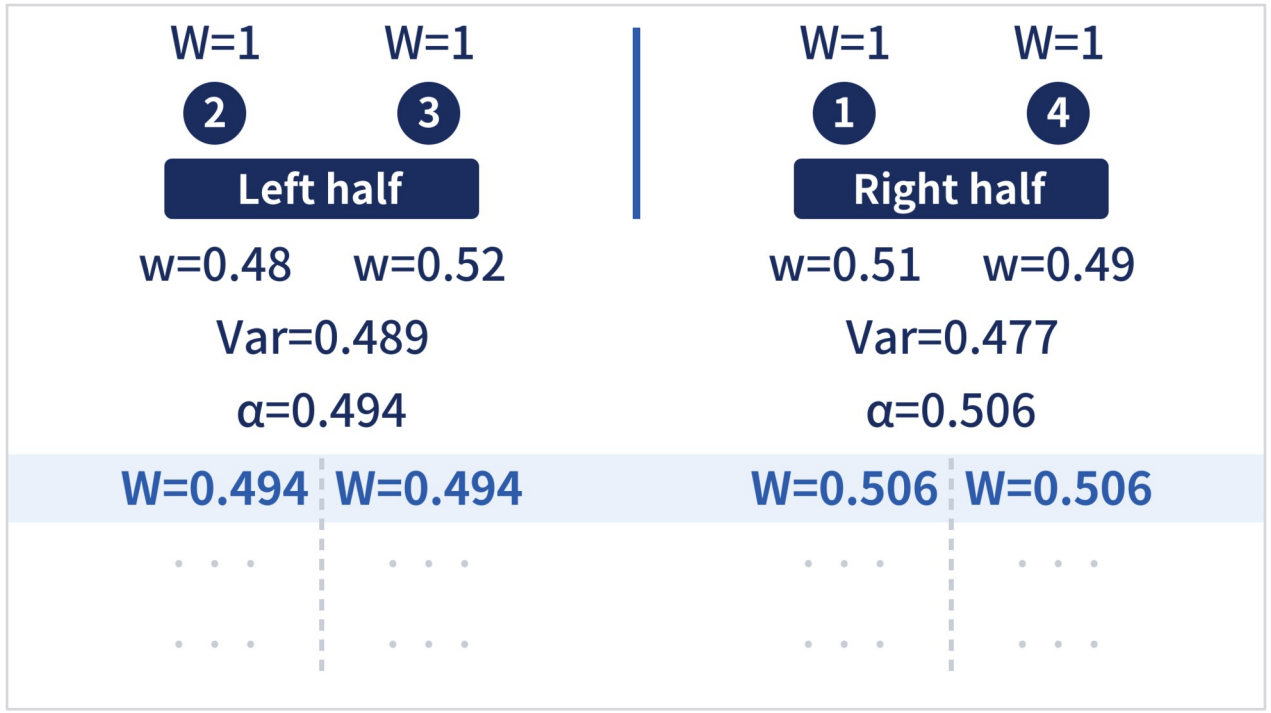
Our numbers (right = 1 − 0.494 = 0.506)

$$\alpha_{\text{left}} = \frac{0.477}{0.489 + 0.477} = 0.494$$

Left gets 0.494 and right gets 0.506, because the right half is calmer

Multiplying by the split ratio finishes level one

Each asset's budget W is multiplied by its half's ratio.



Rule

$$W_i \leftarrow W_i \times \alpha$$

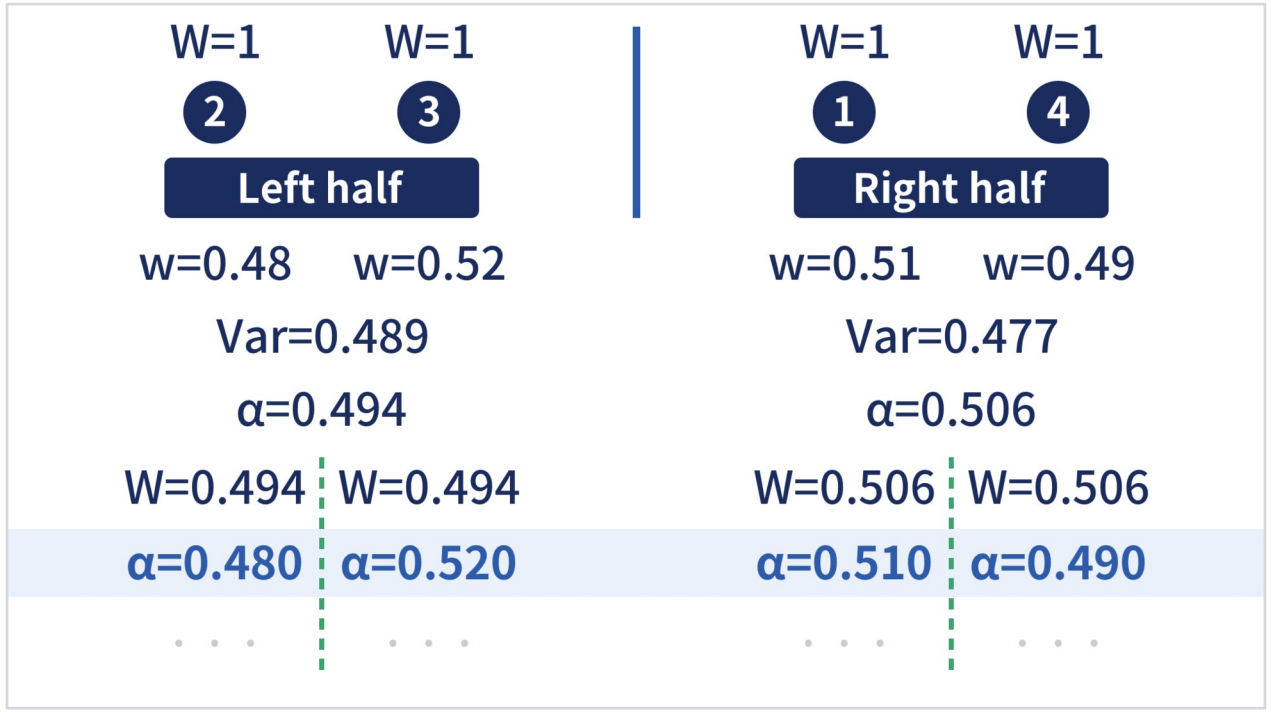
Our numbers

$$W_2 = W_3 = 1 \times 0.494 = 0.494$$
$$W_1 = W_4 = 1 \times 0.506 = 0.506$$

Only the ratio multiplies W; the temporary weights 0.48 and 0.52 are thrown away

With one asset per half, its own variance is used

The same four moves run inside each pair. A single asset has temporary weight 1.



Ratio inside each pair (partner gets 1 minus)

$$\alpha_2 = \frac{0.72}{0.78 + 0.72} = 0.480$$

$$\alpha_1 = \frac{0.77}{0.74 + 0.77} = 0.510$$

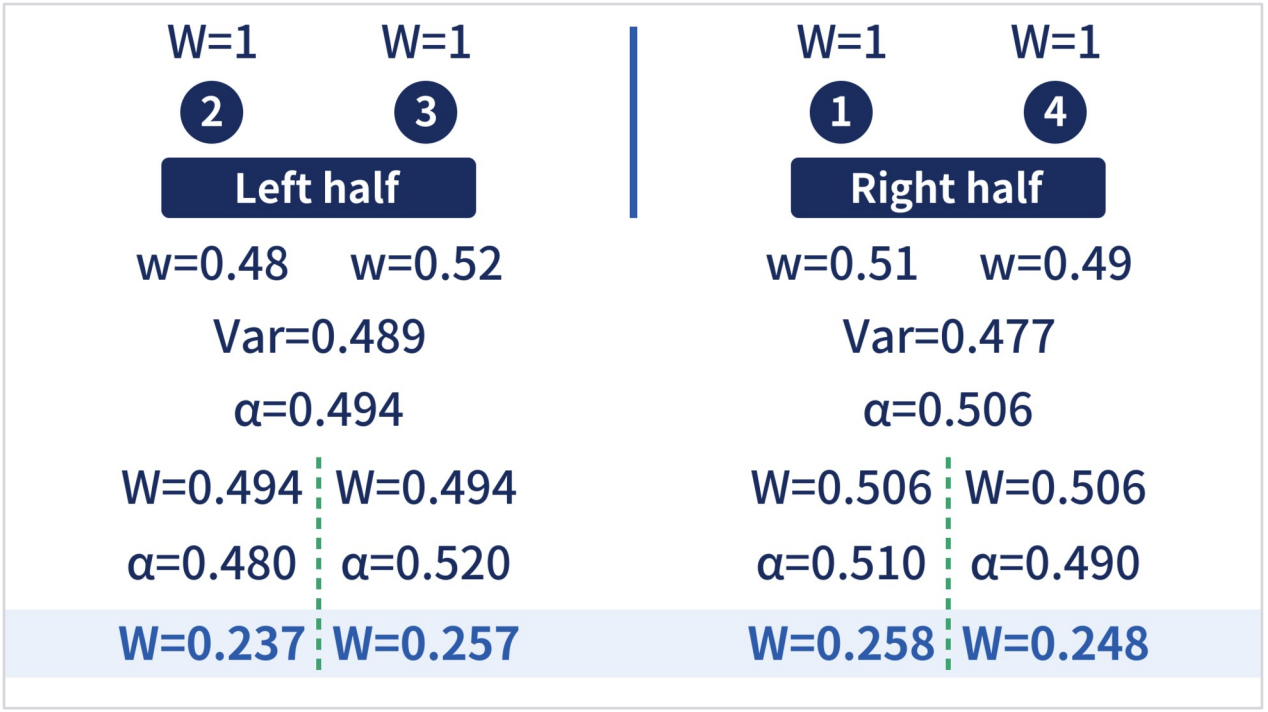
Notice

- Same as Step 2's weights

With only two assets, the split ratio equals the inverse-variance weight

The second split fixes the final weights

Each level-one budget is multiplied by its level-two ratio. The animation rounds to 2 dp.



Final weights

$$W_2 = 0.494 \times 0.480 = 0.237$$
$$W_3 = 0.494 \times 0.520 = 0.257$$
$$W_1 = 0.506 \times 0.510 = 0.258$$
$$W_4 = 0.506 \times 0.490 = 0.248$$

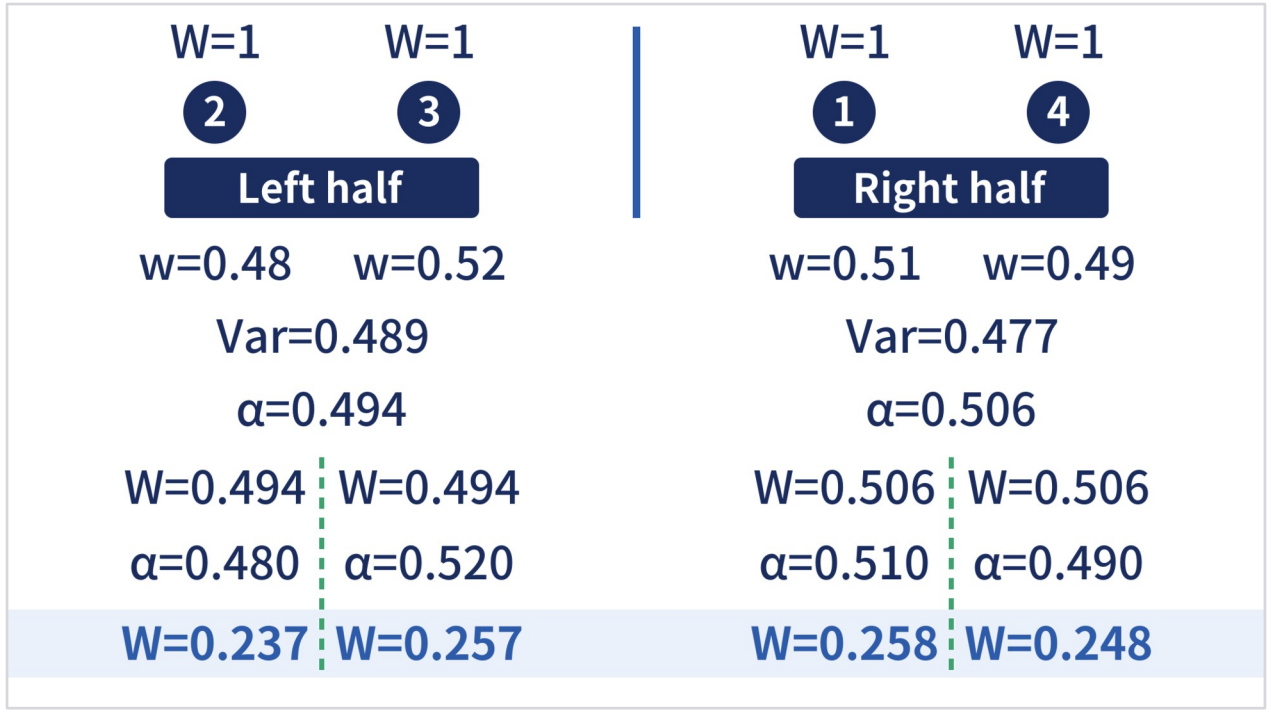
Check: they sum to one

$$0.237 + 0.257 + 0.258 + 0.248 = 1.000$$

Final weights: A1 0.258, A2 0.237, A3 0.257, A4 0.248

The whole calculation fits in one picture

Columns follow the leaf order. The blue row is the answer; dashed lines are the second split.



How to read it

- Level one: between the pairs
- Level two: inside each pair
- w only feeds the variance
- Cross-pair covariance never used

Correlation across pairs shapes the order only; it never enters the weight arithmetic

05

PART 5 · RESULTS

Reading the Results

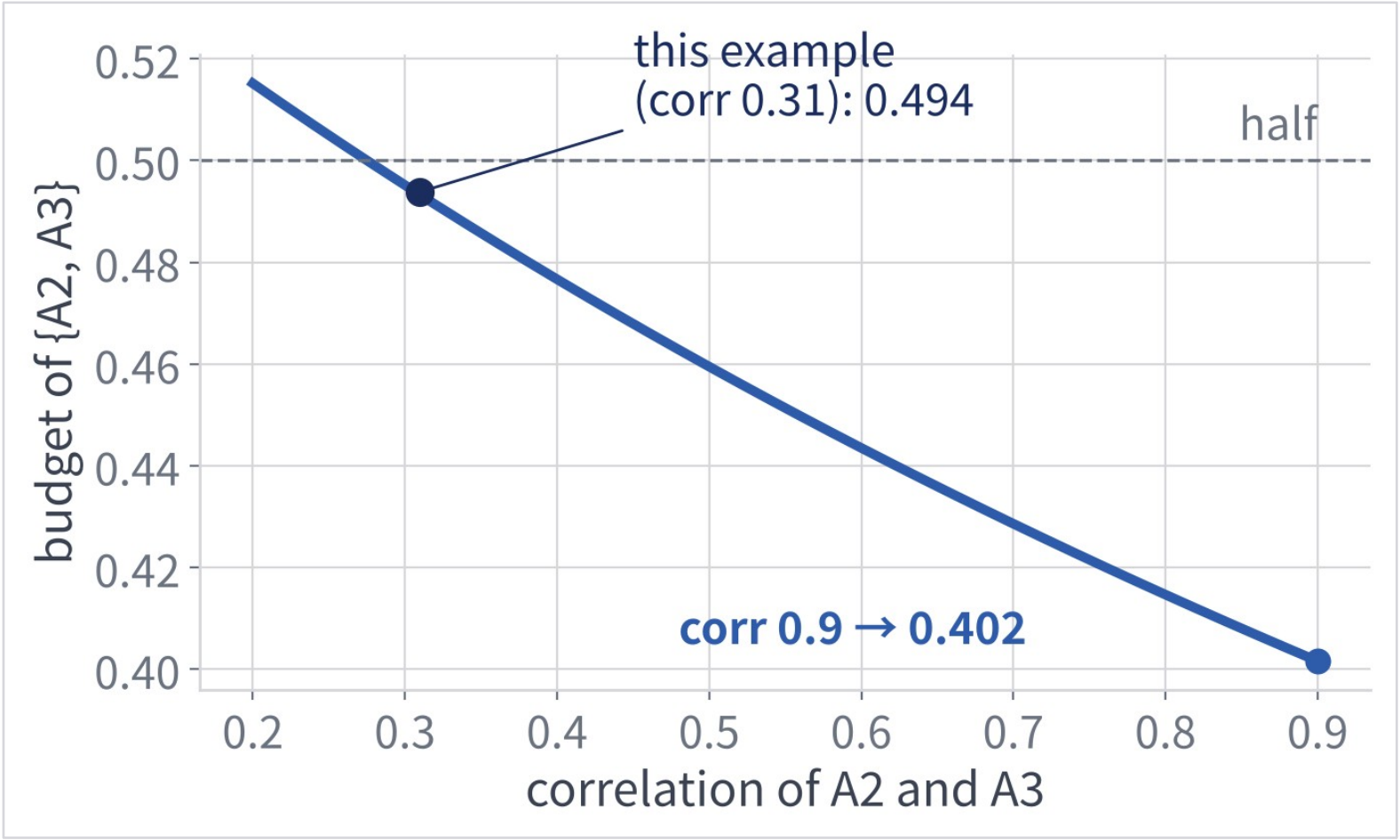
A what-if on correlation, the original 14-asset result, and HRP's strengths and limits.

Alike assets make a group's budget shrink

Only the A2–A3 correlation changes; everything else stays as in the example (illustration).

Why it falls

- Higher corr → riskier half
- Riskier half → smaller ratio
- Corr 0.9 → budget 0.402
- Never drops to zero



On 14 assets, HRP gives every asset a positive weight

Last screen of the MathWorks example: in-sample risk of HRP versus mean-variance.

Reading it

- MVO: assets 7, 8 at zero
- HRP: all 14 positive
- Risk: HRP 0.3982
- Risk: MVO 0.3869

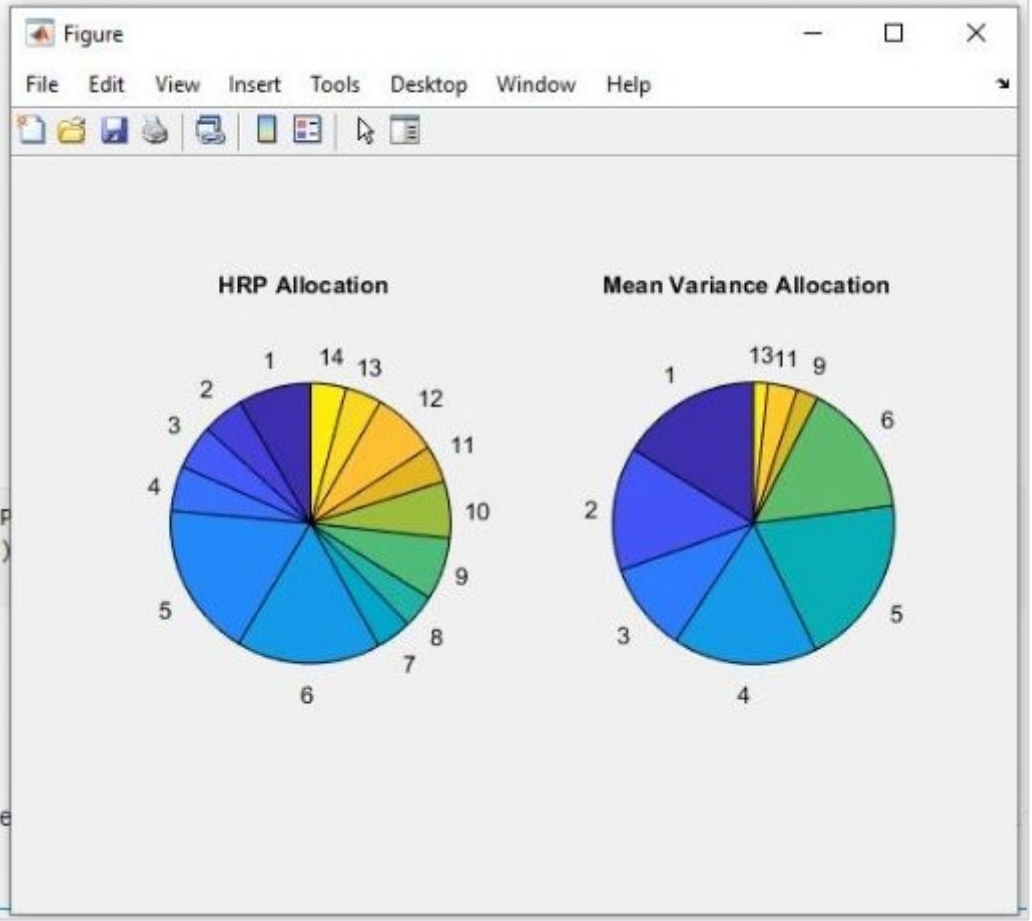
wgtTable = 14×2 table

	HRP	MeanVariance
1	0.0841	0.1622
2	0.0482	0.1420
3	0.0513	0.1028
4	0.0518	0.1662
5	0.1787	0.1972
6	0.1668	0.1543
7	0.0415	0
8	0.0385	0
9	0.0721	0.0250

```
HRP_Risk = sqrt(portvar(assetRetn,wgtHRP
MV_Risk = sqrt(portvar(assetRetn,wgtMV')
table(HRP_Risk, MV_Risk)
```

	HRP_Risk	MV_Risk
1	0.3982	0.3869

You can see that HRP produces a much more dive
and extract their performance metrics.



HRP buys stability by simplifying a few things

Points raised in practice and research since López de Prado (2016).

What HRP does well

- No covariance inversion
- Every asset gets a positive weight
- Reads correlation as a tree
- Simple enough to follow by hand

What to watch

- The halfway cut can split a group
- Cross-group covariance unused
- Expected returns not used
- In-sample risk can exceed MVO

Variants that cut at group boundaries (such as HERC) address the first limit

06

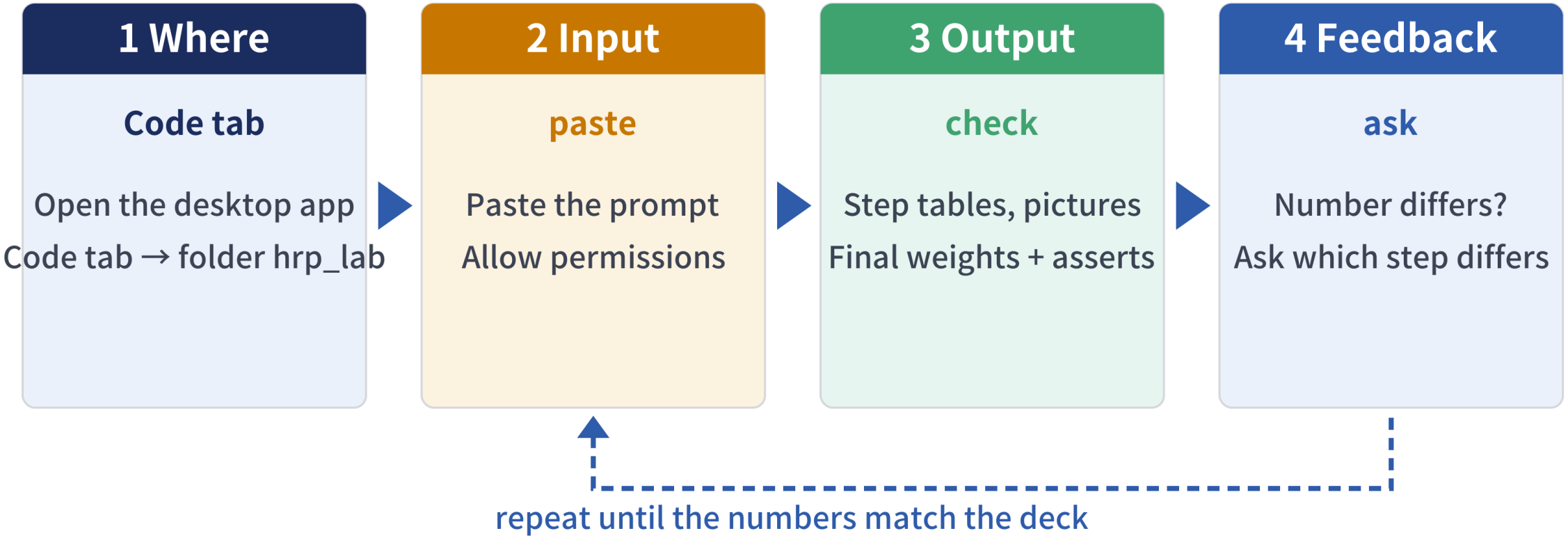
PART 6 · BUILD

Rebuilding It with Claude Code

Paste prompts into the Code tab of the Claude desktop app and check the results.

Students paste the prompts and only check the results

Claude Code installs packages, runs the code and fixes errors; students never edit code.



Prompt 1 rebuilds the four-asset example step by step

Claude Code checks itself against the deck's numbers with assert statements.

[Input] Prompt · Claude Code (Code tab)

```
In the folder hrp_lab, rebuild the four-asset HRP
example in Python.
1) Variances 0.74, 0.78, 0.72, 0.77 (assets 1-4);
   cov(A2,A3)=0.23, cov(A1,A4)=0.20, all others 0.05.
2) Like the MATLAB example: distance ((1-corr)/2)^0.5
   and scipy linkage with 'single' and 'euclidean'.
3) Print the leaf order and, for every bisection step,
   the temp weights, half variances and split ratios.
4) Assert final weights 0.258, 0.237, 0.257, 0.248.
5) Save the dendrogram and heat map as PNG, open them.
   Install packages and fix errors yourself.
```

What to check

- Leaf order 2, 3, 1, 4
- Variances 0.489, 0.477
- Ratios 0.494, 0.506
- Four weights, sum 1

If the asserts pass, the code and the hand calculation agree

Prompt 2 tests HRP against min-variance on ETFs

Claude Code also downloads the data; the prompt rules out any look-ahead.

[Input] Prompt · Claude Code (Code tab)

```
Download daily adjusted prices since 2015 with
yfinance for 14 US ETFs: SPY QQQ IWM EFA EEM VNQ
TLT IEF SHY LQD HYG TIP GLD DBC.
1) Each month end, estimate covariance from the past
   3 years; compute HRP and minimum-variance weights.
   Reuse the HRP function from Prompt 1.
2) Score only with next month's returns (no look-ahead).
3) Table: annual volatility, max drawdown, turnover,
   number of zero-weight assets, for both methods.
4) Plot the last allocations as side-by-side pies.
```

When reading

- HRP: no zero weights
- Where does min-var pile up?
- Compare turnover too
- Odd? Ask which month

Check whether the MathWorks conclusion holds on other data and periods

HRP fixes an order, then halves the budget

1

Order: tree clustering and quasi-diagonalization

Correlation → distance → dendrogram → leaf order 2, 3, 1, 4.

2

Weights: recursive bisection

Split in half, give the safer half more, repeat down to each asset.

3

Result: no inversion, no zero weights

Final weights 0.258, 0.237, 0.257, 0.248 — all close to a quarter.

Next: run Prompt 2 and see how HRP behaves on real data